

Loops

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We often need to repeat actions.

Loops are a way to repeat same code multiple times.

The "while" loop

Syntax:

```
while(<condition>){  
  //code  
  //so-called "loop body"  
}
```

```
const prompt = require('prompt-sync')();  
let numOfTerms = Number(prompt('Enter number of terms: '));  
let i = 0;  
while(i < numOfTerms){  
  console.log(i)  
  i++;  
}  
console.log(`\nValues of j`)  
let j = 3;  
while(j){  
  console.log(j);  
  j--;  
}
```

Output:

Enter number of terms: 5

0
1
2
3
4

Values of j

3
2
1

A single execution of the loop body is called an iteration. The loop in the example above make 5 iterations and second one 3 iterations.

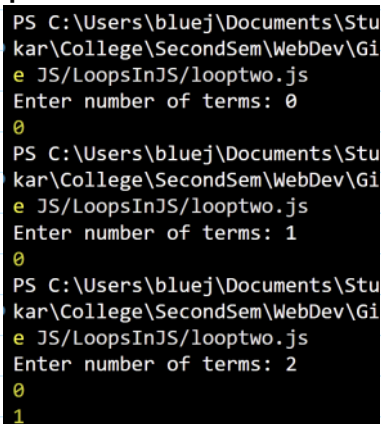
The "do...while" loop

The condition check can be moved below the loop body using the do...while syntax:

```
do{
  //loop body
}while(<condition>);
```

```
const prompt = require('prompt-sync')();
let numOfTerms = Number(prompt(`Enter number of terms: `))
let i = 0;
do{
  console.log(i);
  i++;
}while(i < numOfTerms);
```

Output:



```
PS C:\Users\bluej\Documents\Stu
kar\College\SecondSem\WebDev\Gi
e JS/LoopsInJS/looptwo.js
Enter number of terms: 0
0
PS C:\Users\bluej\Documents\Stu
kar\College\SecondSem\WebDev\Gi
e JS/LoopsInJS/looptwo.js
Enter number of terms: 1
0
PS C:\Users\bluej\Documents\Stu
kar\College\SecondSem\WebDev\Gi
e JS/LoopsInJS/looptwo.js
Enter number of terms: 2
0
1
```

This form of syntax should only be used when you want the body of the loop to execute at least once, regardless of the condition being truthy. Usually the other form is preferred `while(...){...}`

The "for" loop

The for loop is more complex, but it's also the most commonly used loop.

Syntax:

```
for(begin; condition; step){
  //loop body
}
```

For example:

```
const prompt = require('prompt-sync')();
let numOfTerms = Number(prompt(`Enter number of terms: `))
for(let i = 0; i < numOfTerms; i++){
  console.log(i);
}
```

Output:

```
e JS/LoopsInJS/forloopone.js
Enter number of terms: 0
PS C:\Users\bluej\Documents\St
kar\College\SecondSem\WebDev\G
e JS/LoopsInJS/forloopone.js
Enter number of terms: 1
0
PS C:\Users\bluej\Documents\St
kar\College\SecondSem\WebDev\G
e JS/LoopsInJS/forloopone.js
Enter number of terms: 2
0
1
```

Let's examine the for statement part-by-part

Part	Statement	Description
begin	<code>let i = 0</code>	Executes Once Upon entering the loop.
condition	<code>i < numOfTerms</code>	Checked before every loop iteration. If false, the loop Stops.
body	<code>console.log(i)</code>	It runs again and again while the condition is truthy.
step	<code>i++</code>	Executes after the body on each iteration.

That is begin. Executes once and then it iterates after each condition, test body and step are executed.

Suggestion:

If you are new to loops, it could help to go back to the example and reproduce how it runs step by step on a piece of paper.

```
const prompt = require('prompt-sync')();
let numOfTerms = Number(prompt(`Enter number of terms: `))
for(let i = 0; i < numOfTerms; i++){
  console.log(i);
}
console.log(i)
```

Output:

```
e JS/LoopsInJS/forloopone.js
Enter number of terms: 3
0
1
2
C:\Users\bluej\Documents\Students\DeeptaNarayanSarkar
\College\SecondSem\WebDev\Git\Coding\Project0\JS\Loop
sInJS\forloopone.js:9
console.log(i)
      ^
ReferenceError: i is not defined
    at Object.<anonymous> (C:\Users\bluej\Documents\S
tudents\DeeptaNarayanSarkar\College\SecondSem\WebDev\
Git\Coding\Project0\JS\LoopsInJS\forloopone.js:9:13)
```

defined inside the loop block. When JS exit the loop then it will wipe/delete local variable. Therefore, we get Reference error.

Solution of above problem:

```
const prompt = require('prompt-sync')();
```

Output:

The for loop ***begin, condition and step*** are optional, if *condition* is not provided by default JS make it truthy. And the loop becomes infinite, i.e., never ending loop. We have to forcefully stop the code.

Output:

```
const prompt = require('prompt-sync')();
let numOfTerms = Number(prompt(`Enter number of terms: `));
let i = 0;
for(; i < numOfTerms; ){
```

```
    console.log(i);  
    i++;  
}  
console.log(`outside the loop i = ${i}`)
```

Output:

```
e JS/LoopsInJS/forloopone.js  
Enter number of terms: 2  
0  
1  
outside the loop i = 2
```